NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.3a Landfalling hurricanes: AI tracks and ensembles to site peak winds

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 14:37 — 2026-10-16 00:57

Stage conclusion

Project report

B1+B4 landfalling hurricanes: AI tracks and ensembles to station peak winds

Objective

When AI global model tracks (+ NHC official intensity) drive an open parametric hurricane boundary-layer wind field model, can station peak winds reach the level of NHC official forecasts driving the same model; and after adding AI ensembles, can the spread cover ASOS observations? Determine whether it is worth continuing (replacing the wind field model once Liang Hu's 3D boundary-layer model is ready and expanding the number of events). Task brief.

Criteria (Written before launching runs on 10-14; must not be relaxed ex post)

- Primary metric: median absolute error and median bias of the forecast-to-observed ratio of peak gust (or 1 min sustained wind) per station per event; all stations within ±150 km of the landfall point.
- Three chains: (a) best track (HURDAT2 / Extended Best Track including RMW); (b) NHC official forecast track + intensity; (c) AI track (AIWP) + NHC official intensity. Same wind field model, lead times of 24/48/72 h (referenced to landfall time). Chains (b) and (c) use only operationally available information; since OFCL does not report Pc/POCI/RMW, post-event b-decks must not be used (added during 10-14 review as input compliance without altering the criteria); supplementary chain (c′) WeatherNext Cyclones is reported only and not included in the criteria.
- **B1 worth continuing**: Median absolute error of (c) at 48 h ≤ 1.1 × (b), and (c) is no worse than (b) on ≥ 2/3 of events; **simultaneously**, the median bias of (a) is within ±20% (otherwise the problem lies in the wind field model rather than AI, and the conclusion should state "wind field model unqualified" without evaluating AI).
- **B4 worth continuing**: Coverage of station observations by the ensemble 10–90% interval is within 60–95%. B4 is executed only after B1 passes the gate.
- If not met, clearly state where it falls short without relaxing the criteria.

Boundaries (Effective 10-15, tightened by Liang Hu on 10-14)

- Workstation does not run computations or store compute data: wind field calculations, station pairing, and large-file downloads of AIWP/GFS/HURDAT2 must all be executed and stored in Colab sessions or TACC [local path omitted] (CRC group space is at ~95%, so large data prioritizes TACC); the workstation only holds scripts, small results <200 MB per item, figures, and reports; if temporary storage >1 GB is truly needed, report to Node B first.
- Supercomputer usage still goes through ComputeOps resource request + registration, with pre-check and submission separated into two commands; CRC defaults to a single node.
- Current status (verified 10-15 02:00): No computations running on the workstation for this node; total is 44 MB, with no files >200 MB. The live GDEX reads in v2c were decoded in memory at the time and not written to disk. Upon restart, the task brief must be written in accordance with this section.

Current Status

Shelved (10-14, supported by all 6 reviewers). Under the refrozen s2b protocol, B1 did not pass the gate: across 9 homogeneous events and 22 stations at 48 h, the Pangu/OFCL median absolute error ratio is 1.021 (satisfying ≤1.1), but only 5/9 events are no worse than OFCL (requiring ≥6/9); using 10 events as the denominator gives 5/10. In accordance with the sequence, B4 is not pursued. This result cannot claim that AI tracks are overall better or worse.

Why it failed - Both chains underestimate by 35–38% at 48 h. This stems from the pre-registered input assumptions: the initial pressure deficit Δp0 is extrapolated by scaling with OFCL intensity (H08 only uses pressure deficit, not Vmax directly), RMW is frozen at the initialization time (Nicole 200 nmi, Milton 5 nmi), and OFCL intensity itself is biased low. The individual contribution of each factor has not been disentangled. - Track indeed affects station winds, but the direction is inconsistent: Pangu has a closer track on only 4/9 events (median 91 km vs. OFCL 105 km). - Each event has only 1–4 qualified stations, and 34/67 stations dropped out during high-wind periods, biasing the sample toward outer-periphery stations.

What conditions are missing and when to restart: See verdict R1–R4—operationally available size and pressure inputs, or Liang Hu's model; ≥15 evaluable events with ≥3 stations per event; a censoring treatment scheme; switching the primary chain must be pre-registered in advance. Any restart constitutes a new protocol version and does not overturn the current gate verdict.

Assets left for Node B: , v1 wind field and ASOS pairing, 850 hPa tracker and AIWP remote reading, and three-chain results.

Finalized Decisions

- 2026-10-14 v2 preparation stops here: only 13 cases remain after strict station-type and height filtering; do not freeze or run (a), wait for restart triggers →
- 2026-10-14 after v2b verification returned: complete prerequisites and produce a finalized pre-registration candidate, still without running the gate evaluation →
- 2026-10-14 after v2 feasibility returned, do not open the v2 gate; dispatch a targeted verification of the hourly homogeneous pool →
- 2026-10-14 B1 failed to pass the gate, node shelved, B4 not pursued; retract the "track signal swamped" attribution; rewrite restart conditions as R1–R4 →
- 2026-10-14 Colab policy correction (Node B Lead, following Liang Hu's exact words): no single-experiment cap, report first if projected >100 CU; must stop upon completion, and dispatch only one executor per round
- 2026-10-14 Phase 2 protocol revision s2b: OFCL 12 h interpolation, homogeneous sample (Nicholas 48 h non-evaluable), primary (c) uses Pangu, (c′) does not use Liang Hu's credentials; criteria unchanged →
- 2026-10-14 Chain (a) v1 passed the ±20% gate; accept the ~2 min ASOS proxy definition; dispatch Phase 2 →
- 2026-10-14 Literature review verdict: OFCL lacks Pc/POCI/RMW, so (b) and (c) must not use post-event b-decks; check CIRA/Weather Lab first for AI tracks, and use 850 hPa circulation for self-tracking; add (c′) WN-C; narrow incremental contribution scope to "AI tracks + classical parametric boundary layer → US ASOS three-chain comparison, not found within searched scope" →
- 2026-10-14 Hierarchy rule (set by Root Lead): this node is Level 2 and does not open Level 3 sub-projects; if B4 needs to be independent, report to the Lab Lead for Node B to open a peer-level node; tool bots must not re-delegate
- 2026-10-14 B1 and B4 opened jointly (set by parent node) →

- 2026-10-14 Phase 1 minimal model (+50.7%) judged as an incomplete implementation; dispatch Phase 1b to retest the same gate with the full model under pre-registration →
- 2026-10-14 Step 1 split into two phases: first "data access feasibility + event list + wind field model implementation + chain (a)", then proceed to (b) and (c) only after chain (a) passes the ±20% gate →

What Was Tried and Results

Date	What Was Done	One-Line Result
10-14	Grok quick search (b99be0) + Lead check	DeMaria 2025 abstract verified: AI tracks ≈ operational, intensity systematically weak (most pronounced in first 24 h) → use NHC for intensity; AIWP README has no TC track files → requires self-tracking; no precedent found for "AI track → parametric wind field → station wind" (Grok only, not yet corroborated by Deep Research)
10-14	Phase 1: Minimal Holland + constant 0.85, chain (a) 5-event trial	4 events and 17 stations after QC, median absolute error 50.7%, median bias +50.7%, failed to enter ±20%; three Idalia stations dropped out before landfall; treated as proxy screening, not a formal gate
10-14	Phase 1b: v1 pre-registration (CLIMADA H08 + roughness + averaging-time conversion), chain (a) 12 events	10 events, 29 stations: sustained wind bias +9.9% (passed gate), median absolute error 21.0%; gust bias −0.9%, median absolute error 11.2%; 34/67 stations censored
10-14	AIWP remote slicing (Milton, FOUR, Colab CPU)	Landfall region msl/u10/v10/u850/v850 totaled 88 MB, 337 requests, 39 s; simple MSLP minimum tracking for demonstration only
10-14	Phase 2: Chain (b) OFCL (s2a interpolation fix, exploratory); Pangu Milton 48 h trial run	(b) 24 h: 10 events, 29 stations, sustained wind bias −5.1%, median absolute error 29.3%; 48 h: 9 events, 22 stations, bias −37.9%, median absolute error 41.1%; Pangu run for Milton single case took ~0.95 GB and 7.8 min, solely to prove end-to-end execution; censored lower bounds: model peak fell below pre-dropout observed lower bound at 2/19 stations for (a) and 11/19 stations for (b) 48 h; (c′) required login, not retrieved
10-14	Phase 2b: s2b protocol, three-chain 48 h gate evaluation	Homogeneous 9 events, 22 stations: OFCL median

Date	What Was Done	One-Line Result
10-14	v2 feasibility check (counting events only, no pairing)	7 pre-screened candidates under 1-minute definition, 14 potential under hourly definition; R1 route exists, environmental field and Holland fitting not closed; do not open v2
10-14	v2b targeted check of hourly homogeneous pool	Exactly 15 events × ≥3 stations in 2020–24 (zero margin); Isaias has no Pangu; GFS environmental field verified for 12/15
10-14	v2c prerequisite completion (station type/height, environmental field, operational timing)	After strict station type and height filtering, 13 events × ≥3 stations, short of 15; 2020 environmental fields retrievable; H08 adapter missing
10-14	GitHub repository table (live API check)	CLIMADA (480★, GPL-3), TCRM (99★, includes Kepert BL), TempestExtremes (151★, BSD-2, tracker), TCHazaRds, ParaTC, weathernext (Apache-2.0, backup for self-running B4 ensembles)

Parameter Source Table (Shared across Node B)

Parameter	Value	Source or "Assumption"	Where Checked
Holland B	Inverted at each step from Vmax, Pc, POCI, and RMW; sample values ~2.1–2.8, bounded to 0.5–4	Assumption: Holland gradient-wind pressure-deficit relation (Wang et al. 2020 PDF p. 5 Eq. 7); taking Vmax/0.85 as gradient wind is an assumption of this experiment	Checked HURDAT2, ATCF b-deck, Vickery 2009 PDF pp. 23–24; no direct B observations obtained
RMW	Column 19 of b-deck, 6-hourly linear interpolation, nautical miles converted to meters	Data source: NHC ATCF b-deck; CIRA EBTRK only updated through 2021, HURDAT2 itself has no RMW	Checked CIRA EBTRK page and README, per-storm b-decks; not imputed when missing
POCI	Column 17 of b-deck, 6-hourly linear interpolation	Data source: NHC ATCF b-deck, not arbitrarily set to 1010 hPa	Checked b-deck structure and EBTRK documentation
Gradient-to-surface wind factor	0.85 (uniform over sea and land)	Assumption/simplification; Vickery 2009 PDF pp. 23–24 states 0.85 is a tuned comparison baseline on the same sample batch, by no means its full boundary-layer model; sensitivity range 0.65–0.90 not completed	Checked card WTPZS77N and original text; its slab/roughness transition was not implemented

Parameter	Value	Source or "Assumption"	Where Checked
Air density	1.15 kg m^-3	Assumption (range 1.10–1.25; observed temperature and humidity not used)	Checked Wang et al. 2020 PDF pp. 4–5; no thermodynamic field in this experiment
Sea/land roughness	Not yet in model (Vickery examples: sea 0.0013 m, land 0.03 m)	Explicit omission, must not claim exposure correction was performed; source card WTPZS77N, Vickery 2009 PDF pp. 12, 15, 22	Checked card WTPZS77N and original text; directional land cover at stations not retrieved
Gust factor / averaging time	Not in primary metric: ASOS sknt serves as ~2 min sustained wind proxy, gust_sknt as 5 s peak; not treated interchangeably as 1 min / 3 s	Candidate card ZR7S3SEJ PDF pp. 2–3, 5–6 documents different averaging durations and roughnesses, candidate card 2KXRIJJB PDF pp. 2–4 specifies ~2 s conditions; no qualified station-specific conversion available	Checked two cards pending independent review and IEM 1min fields; no uniform gust multiplier assumed
v1 CLIMADA H08	CLIMADA 6.1.0, commit bb9595944fcf673152ea53e61efbafeb0b1e4406; native 10m 1min Holland 2008 + translation vector	Open-source implementation lines 1131–1257; not Vickery slab, 0.9 default reduction factor in source lines 61–69	Checked GitHub pinned tag, actual runtime package, and v1 pre-registration
v1 directional exposure	Natural Earth 10m land polygons; land/sea majority vote across three points at 2/5/10 km upwind along model inflow; z0 land 0.03m / sea 0.0013m; logarithmic profile ratio with 500m reference height	z0 source: card WTPZS77N, Vickery 2009 PDF pp. 12/15/22; 500m reference height and single-class surface are assumptions (range 200–1000m, land z0 0.02–0.07m)	Checked Vickery card, Natural Earth 10m land; CLIMADA does not include directional station exposure correction
v1 averaging time	1min→~2min sknt ×0.97; 1min→5s gust_sknt land ×1.30, sea ×1.20	All are assumptions, ranges 0.95–1.00, land 1.15–1.45, sea 1.10–1.35 respectively; Vickery and Skerlj 2005 candidate card ZR7S3SEJ PDF pp. 2–3/5–6 provides averaging definitions/exposure basis without directly giving the above factors	Checked candidate cards ZR7S3SEJ and 2KXRIJJB (1992 PDF pp. 2–4 ~2s/10min, cannot be applied directly); v1 was not tuned against observations
s2 (b)(c) initial size and ambient pressure	Persistence of CARQ τ=0 RMW/POCI; if absent, same-cycle OFCL τ=0; if both absent, cycle is marked missing	Operationally available data plus persistence assumption, strictly no post-event b-deck	Checked NHC ATCF a-deck cycle by cycle and review verdict; see

Parameter	Value	Source or "Assumption"	Where Checked
s2 (b)(c) future central pressure	Δp(t)=Δp0×[OFCL V(t)/V(0)]^(1/0.644); constant POCI; CARQ initial pressure prioritized	Exponent borrowed from Atkinson–Holliday 1977 (Western North Pacific), Atlantic anchor scaling is an assumption, not literature-validated	DTIC abstract and Knaff–Zehr 2007 text restatement; for missing-value fallback see
s2 AI 850hPa tracking	5×5 ζ smoothing, mean within 200 km circle, 350 km search radius from previous center, and cyclonic circulation >0	Specific thresholds are all assumptions, tracking direction follows DeMaria 2025 PDF pp. 14–15; not an exact replica of its Table A2	Checked original appendix and ; MSLP used as auxiliary only

Open Items and Next Steps

- Shelved. v2 materials are parked at (PREREG_v2 candidate unfrozen). Restart triggers: ① Liang Hu's 3D boundary-layer model becomes available (replacing H08 while resolving the B adaptation issue); or ② the 2026 season enters HURDAT2, AIWP fills in the Isaias cycles, and historical anemometer heights are verified via HOMR, yielding ≥15 events × ≥3 stations. Upon restart, first decide whether to include 7.9 m (26 ft) ASOS stations with height correction (including them gives 3 stations each for Delta and Milton, reaching 15 events, though with zero margin), which must be written into the pre-registration before inspecting any errors
- 10-14 v2 feasibility check returned (, unfrozen): only 7 pre-screened candidates under the 1-minute ASOS definition; 14 potential under the hourly definition (Eta unverified); 1-minute and hourly peaks cannot be pooled together; R1 selects OFCL wind radii + Knaff 2007 RMW prior + Knaff–Zehr 2007 ΔP (requiring GFS V500/Penv), where Holland inverse fitting is an engineering assumption. Conclusion: **do not open the v2 gate now**, remain shelved
- 10-14 20:4x dispatched targeted check (queued): recalculate station counts within 150 km using only hourly-reporting stations (previous 11 + Eta/Elsa/Fred/Henri + 2026 season ≥50 kt landfalls), and check whether operational GFS archives can compute V500/Penv; task brief; 0 CU. Only draft and freeze v2 if ≥15 events × ≥3 stations are reached and environmental fields are available
- 10-14 21:54 targeted check returned: under the single hourly METAR definition, 2020–24 has **exactly 15 events × ≥3 stations** (170 station-events), all with matching-cycle Pangu and OFCL, leaving a margin of 0; the 2026 season has only Isaias as a ≥50 kt landfall, but without Pangu objects; GFS f048 environmental fields are available for 12/15 cases, with three 2020 cases returning AWS 404; V500 currently uses 10 m wind, pending level verification against the original paper
- 10-14 22:0x dispatched (queued): fill in the three 2020 environmental fields, verify the V500 level, set operational timing rules per the NCEP release schedule, verify ASOS/AWOS status and anemometer heights station by station before recounting events, check Isaias, and deliver a freezable (hourly definition as primary, chain (a) re-passing ±20% under the same definition, no gate evaluation if <15×3 after QC); task brief; 0 CU. Lead will freeze upon return, then run (a) only

- 10-14 22:20 prerequisite completion returned: the three 2020 GFS environmental fields are computable via live reads from NCAR GDEX (V500 switched to 850 hPa following Knaff et al. 2014, as the original KZ07 paper did not specify the level); operational timing uniformly steps back one cycle to take f054 according to the NCEP release schedule (acting as a proxy); based on the conditional snapshot of the NWS 2025 ASOS equipment list with anemometer heights of 8–12 m, only **13 events × ≥3 stations** qualify (Delta and Milton have 1 station each), and historical heights are not verified via HOMR; the Pangu cycles for Isaias are a data void; B in CLIMADA H08 is computed endogenously so wind radii can only fit RMW, and the adapter is neither written nor validated. **Lead decision: stop v2 preparation here, do not freeze, and do not run (a)**; see decision
- The pre-registered item "observation window completeness at the time of the new model peak" was not delivered and is retained as an interpretive limitation

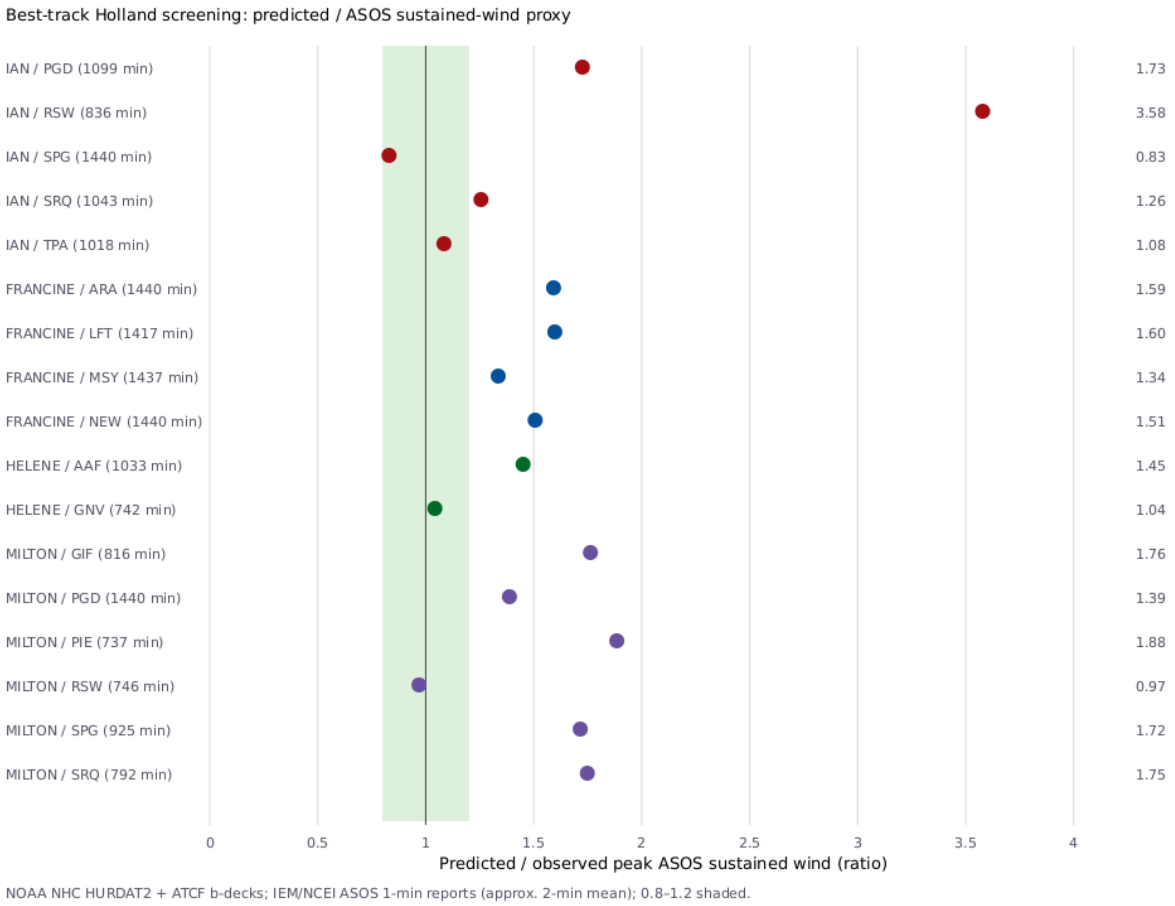
Literature Basis

- 2026-10-16 data: 7,524 NOAA HRD dropsonde 10–300 m typhoon wind profiles (Ma et al. 2026, Zenodo 10.5281/zenodo.21868265, CC BY 4.0; approved X2 by Liang Hu with instructions to archive) → (backed up on CRC). Useful for profile shape, 10–300 m wind shear, comparison against H08/empirical profiles (over ocean), and profile constraints if B1+B4 is restarted; not overland profiles. Does not alter this node's status or gates.
- (10 height-stratified model cards)
- (Vickery 2009 boundary-layer / near-surface conversion)
- (Hurricane Gust Factors Revisited), (Gust factors applied to hurricane winds): gust factors / averaging-time conversion
- (DeMaria 2025): AI tracks are comparable to GFS/ECMWF and can serve as guidance (PDF pp. 5–8, 11–13); intensity is almost always biased low, with mean absolute error ~40 kt after 72 h and no intensity skill (PDF pp. 6–9, 13) → chain (c) in this node uses NHC intensity; splicing "AI track + external intensity" was not tested by the authors and represents an engineering inference by this node (PDF pp. 1, 11–13); AI track files are not publicly released (PDF p. 14); not a landfall-specific sample
- (Radford 2025, AIWP dataset)
- R1 input method cards (10-14): (wind–pressure relation Eq. 7, also requiring ϕ, c, Penv, V500; PDF pp. 4–5, journal pp. 74–75), (Rankine x is not equal to Holland B), ; index. OFCL Vmax and wind radii alone cannot close the Holland parameters; analysis fields or AI fields are needed to supply Penv and V500
- Neighboring precedents (10-14 litwiki cards created, all distinct from this node: none is "AI track + parametric boundary layer → US ASOS station peak winds"):
- : NHC wind speed probabilities, generating 1000 members from official forecasts plus error distributions using the radii-CLIPER wind field; verified against best-track wind radii grids rather than station peaks (PDF pp. 11–12; journal pp. 1583–1584)—operational baseline for B4
- : adds GPCE stratification, improving Brier score by 1%–4% (PDF p. 11; journal p. 596)
- : WN-C with at least 50 members; wind speed probabilities evaluated using IBTrACS wind radii; the paper explicitly states "per-location winds are left for future work" (PDF p. 7; journal p. 686)—direct positioning of this node's incremental contribution

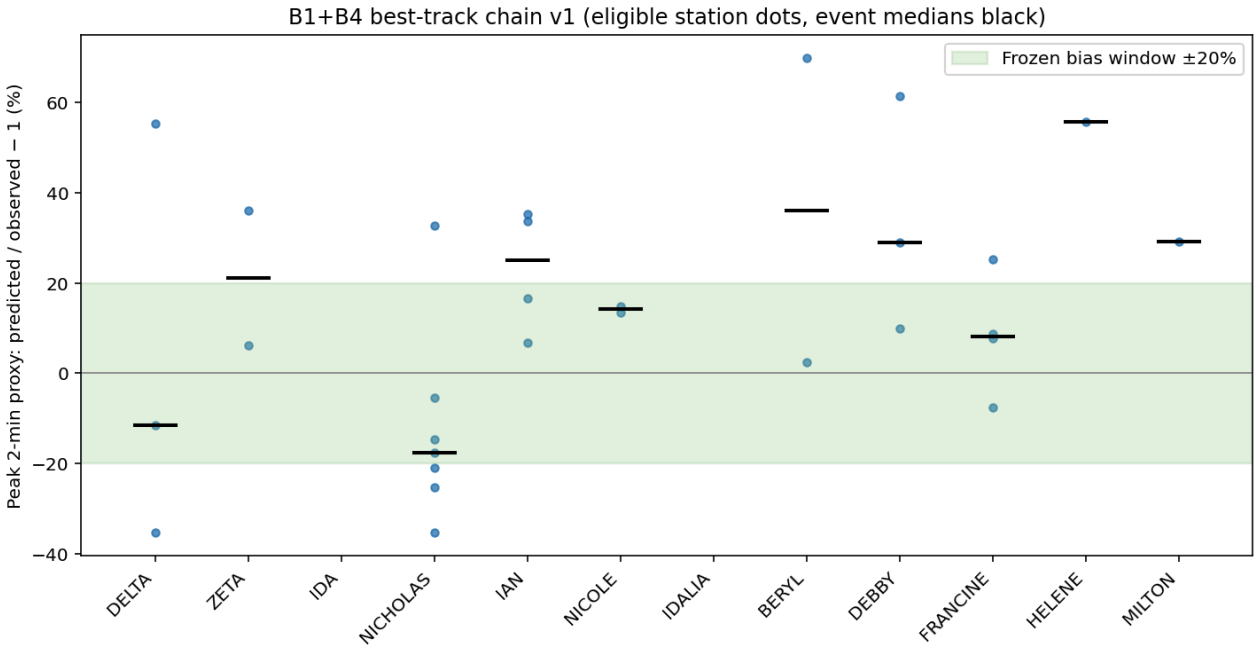
- : Pangu/AIFS driving three-way coupled WRF (7 TCs affecting Hong Kong), reducing track error by 28%–34% compared to IFS-driven runs; no station peak wind verification performed (PDF pp. 3, 10)
- : Pangu gridded wind bias correction plus downscaling across 2513 stations in Zhejiang, reducing 2-min hourly wind MAE by 38.8%; does not use a parametric wind field and does not evaluate event peaks (PDF pp. 12–13)
- Radford et al. 2025 BAMS 10.1175/BAMS-D-24-0057.1 (AIWP data; already in PUBLIC_CITATIONS)

Liang Hu's Notes

Key figures

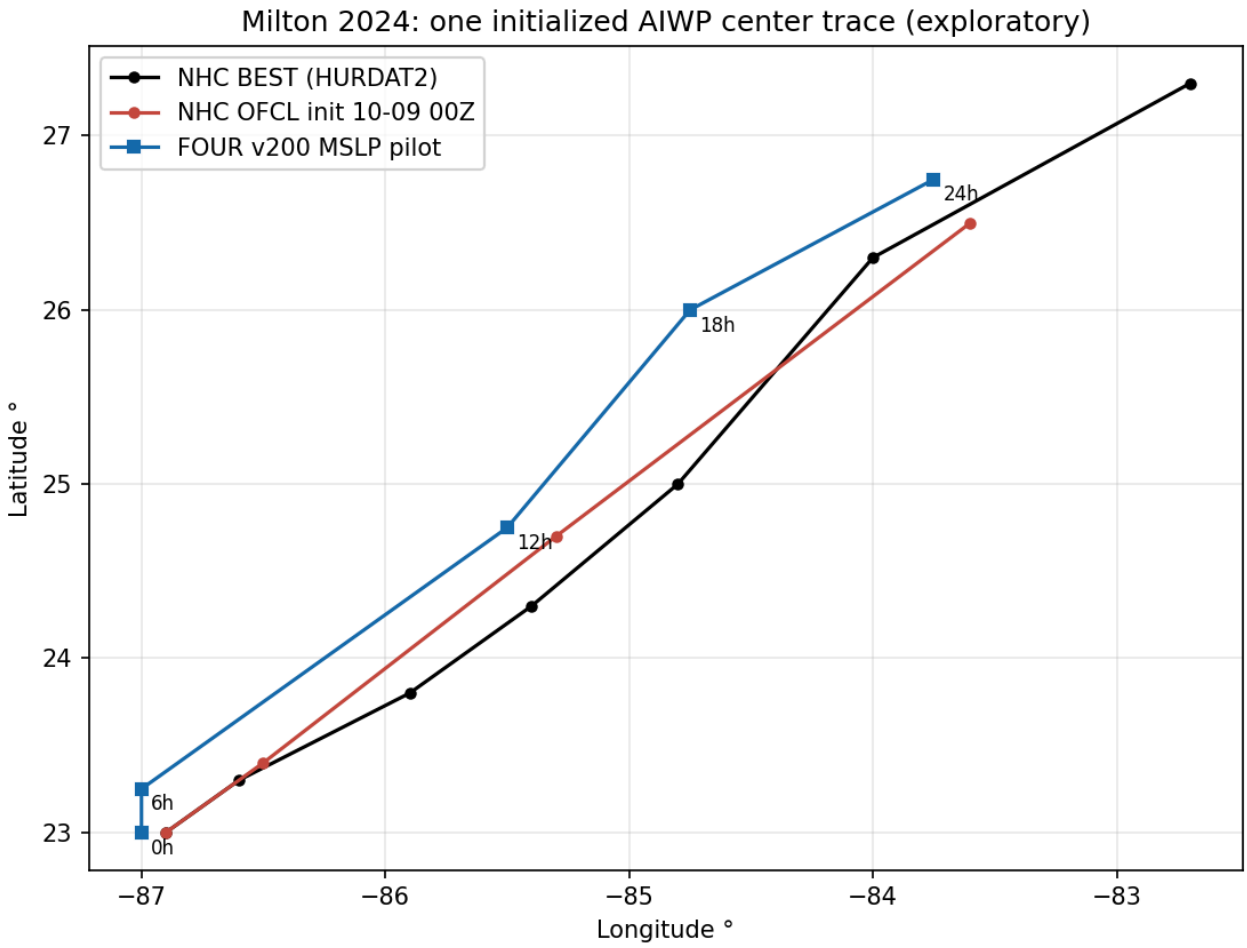


chain a ratio pilot5 v2



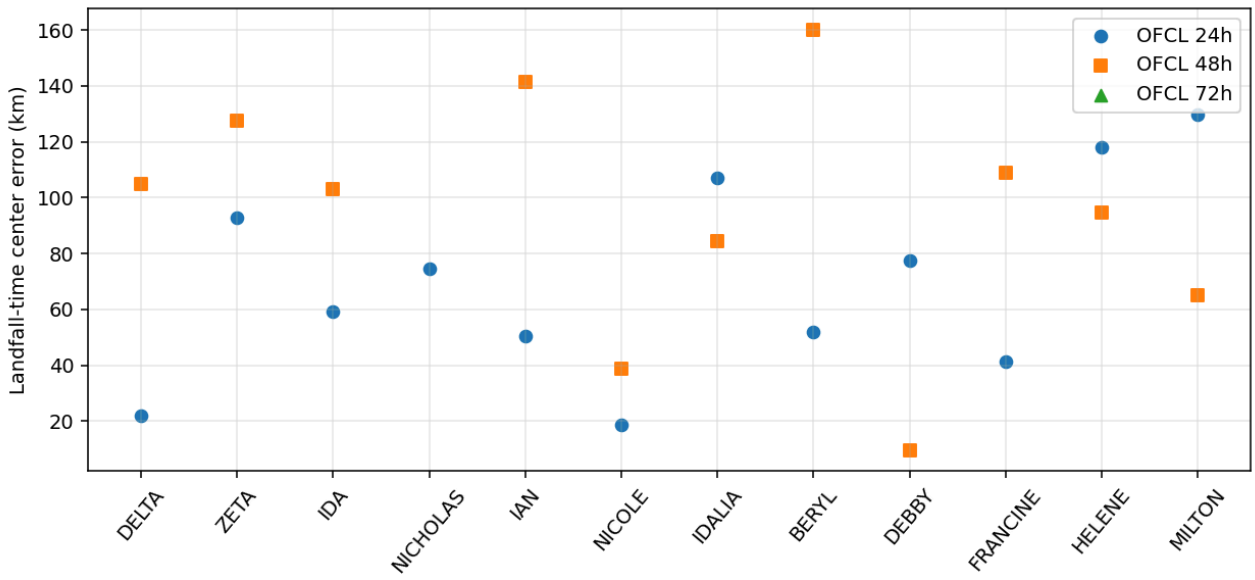
Sources: NHC HURDAT2 + ATCF b-decks; IEM/NCEI ASOS 1-min; Natural Earth 10m land; CLIMADA v6.1.0.

chain a v1 ratio



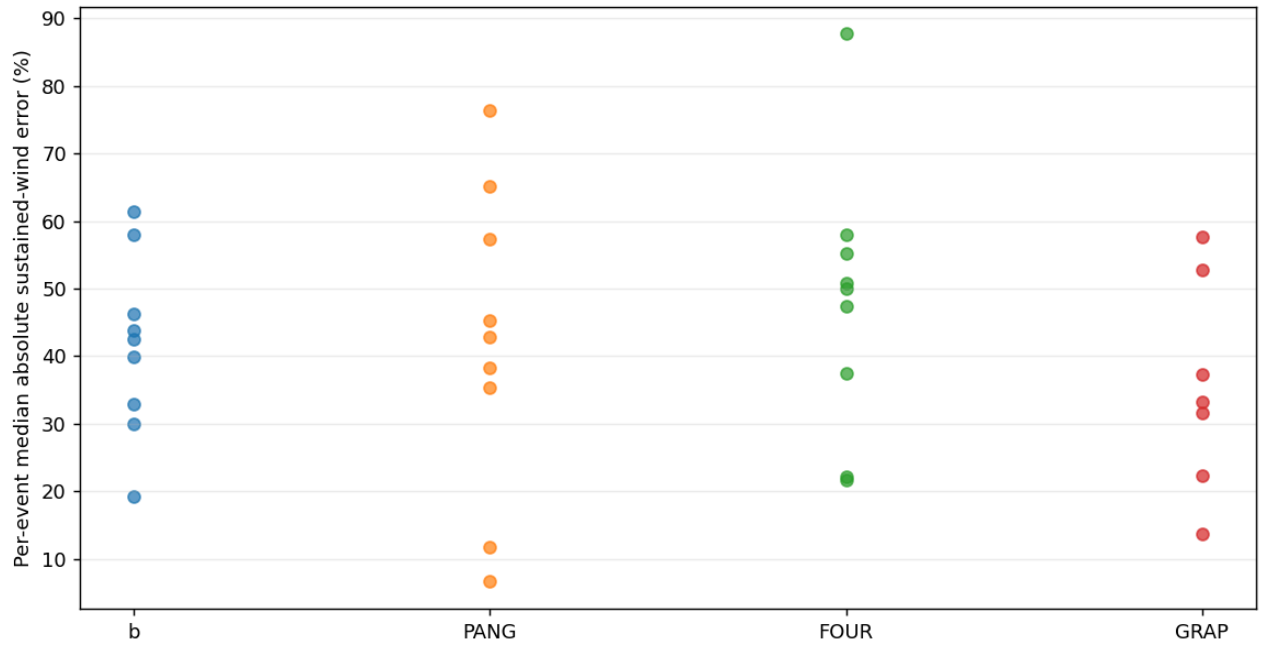
Sources: NOAA/NHC HURDAT2 & ATCF a-deck; CIRA/NOAA AIWP FOUR v200 GFS, 2024-10-09 00Z.

aiwp milton path pilot



Source: NOAA NHC ATCF a-deck OFCL and HURDAT2 BEST; protocol-corrected s2a (exploratory)

b path errors s2a



Sources: NOAA NHC ATCF OFCL / NOAA AIWP Pangu, FOUR, GRAP; IEM ASOS fixed eligible stations. s2b 48h.

s2b event errors